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Gaussian Difference: Find Any Change Instance in 3D Scenes

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***Abstract*—Instance-level change detection in 3D scenes presents significant challenges, particularly under uncontrolled conditions without labeled image pairs, varying camera poses, or restricted lighting. This paper addresses this challenge by developing a novel approach to detect changes in real-world scenarios. Leveraging 4D Gaussians to embed multiple images into 3D Gaussian distributions, our method enables the rendering of two coherent image sequences. By segmenting each image and assigning a unique identifier to each instance, we can efficiently identify changed instances through ID comparison. Additionally, we utilize change maps and classification encodings to categorize the 4D Gaussians as changed or unchanged, allowing for the rendering of a comprehensive change map from any view direction. Through extensive experiments on various instance-level change detection datasets, our method demonstrates significant improvements in detection accuracy over state-of-the-art methods like C-NeRF and CYWS-3D, particularly in scenarios with large lighting variations.**

***Index Terms*—Change detection, 4D Gaussian, instance segmentation, 3D.**

I. Introduction

Most existing change detection (CD) methods are based on coupled images and detect changes according to the features of the 2D appearance, which have been extensively researched in various domains, including street-view scenes [1], [2] and remote sensing [3]–[5]. Lei et al. [6] propose a hierarchical paired channel fusion network to improve the accuracy of change maps. Wang et al. [7] propose an effective multi-level feature interaction method to achieve high-performance change detection. However, these CD methods heavily rely on large-scale labeled datasets (e.g., [4], [8], [9]) that require precise camera pose alignment, which can be a significant challenge.

In contrast to conventional CD methods, 3D CD is a technique that aims to identify change objects of a scene over time using the geometric information of the 3D scene, which can generate a change map of an arbitrary viewpoint [10]–[13]. Typically, 3D CD takes multi-view optical images and depth images before and after scene change. Fehr et al. [14] develop a 3D reconstruction algorithm that utilizes an extended Truncated Signed Distance Function to more accurately solve the scene differencing problem. Ku et al. [15] have contributed a street-scene dataset for 3D point cloud change detection that can detect changes in a complex street environment from multi-temporal point clouds. Qiu et al. [16] propose a method to explicitly localize changes in 3D bounding boxes from two point clouds. Sachdeva et al. [17] propose a method for change detection by using depth maps to map the images before and after changes into point clouds. Achieving high-quality 3D point clouds, particularly from real-world scenes, can be both costly and time-consuming. Recent research [13] introduced a change detection method called C-NeRF, based on Neural Radiance Fields (NeRF) [18], which enables change mapping using only two image sequences captured from the scene before and after the change occurs. C-NeRF constructs two aligned NeRFs and renders image pairs to calculate the absolute difference for identifying changes. While it can render a high-quality change map from new viewpoints, it struggles with scenes that have significant lighting variations and requires lengthy processing time.

In this paper, we focus on designing a robust and efficient instance-level CD method in 3D scenes. Our approach is designed to handle scenarios with uncontrolled lighting conditions while ensuring fast processing times and eliminating the need for manual threshold setting. To achieve this, we propose *Gaussian Difference* which builds on 4D Gaussian Splatting [19] that uses a deformation field to model temporal changes of Gaussian distribution on the basis of 3D Gaussian [20]. 4D Gaussian effectively captures multi-time stamp information of the scene using a single model. We embed multiple images captured before and after the scene change into 4D Gaussian. By interpolating camera poses, we generate a smooth image sequence composed of rendered images from both time points. We segment each image with SAM [21] to obtain independent instances and track the instance with DEVA [22] to endow each instance with a unique identifier. By comparing the IDs in the image pairs, we can efficiently identify the change instances. The change maps are used to retrain the 4D Gaussians with newly added classification encodings, enabling the partitioning of Gaussians into 'changed' or 'unchanged' categories. Gaussian Difference can accurately render change maps from any viewpoint. We have conducted abundant experiments on various scenes. Compared with C-NeRF [13] and CYWS-3D [23], Gaussian Difference demonstrates comparable detection performance on the scenes with uniform lighting and significant improvements in detection accuracy on the scenes with large lighting variations. Our main contributions are as follows:

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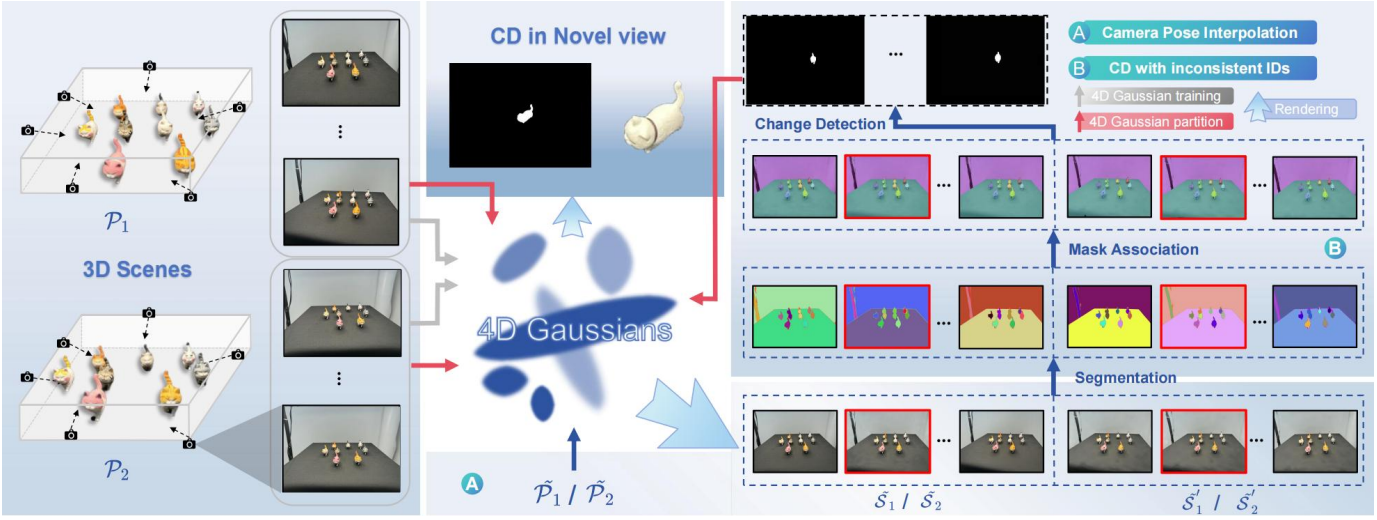

Fig. 1. The main framework of the proposed 3D change detection method.

- We propose Gaussian Difference, a 4D Gaussian Splatting-based change detection framework, which is capable of generating a change map of any view direction.
- Gaussian Difference is efficient in training and robust to the lighting changes of scenes and does not require manually setting the threshold for each scene.
- We enrich the 3D instance-level CD dataset with more scenes that have large lighting variations, which will be released to boost the study of 3D change detection.

II. METHODOLOGY

The purpose of our work is to establish a 3D scene representation of the change instances, which can accurately render the image of the changes from novel viewpoints. As shown in Fig. 1, our 3D representations are built on 4D Gaussian splatting [19]. Given two image sets $\mathcal{S}_1 = \{\mathbf{I}_1^1, \mathbf{I}_2^1, \ldots, \mathbf{I}_{N_1}^1\}$ and $\mathcal{S}_2 = \{\mathbf{I}_1^2, \mathbf{I}_2^2, \ldots, \mathbf{I}_{N_2}^2\}$ captured before and after change, we initially learn a 4D Gaussian representation Ψ. And then we interpolate poses to generate smooth variations from $\mathbf{I}_1^1$ to $\mathbf{I}_{N_1}^1$ and $\mathbf{I}_1^2$ to $\mathbf{I}_{N_2}^2$, which can be used to generate consistent IDs of the instances across different frames. By comparing the IDs of the two scenes, we can easily identify the change instances and their corresponding masks through track-based instance-level change detection. Finally, we propose 4D Gaussian distribution partitioning to divide the Gaussians into changed Gaussians Ψ_{c} and unchanged Gaussians Ψ_{uc} with the change maps. After partitioning, we can generate a change map of any given viewpoint.

A. Camera pose interpolation

Let $\mathcal{P}_1 = \{p_1^1, p_2^1, \ldots, p_{N_1}^1\}$ and $\mathcal{P}_2 = \{p_1^2, p_2^2, \ldots, p_{N_2}^2\}$ represent the camera poses of the images in $\mathcal{S}_1$ and $\mathcal{S}_2$, respectively. Note that $p_i^k = (Q_i^k, T_i^k)$ denotes the camera pose of the i-th image of the k-th image set. Q_i^k and T_i^k denote the rotation quaternion and translation matrix, respectively. We combine $\mathcal{P}_1$ and $\mathcal{P}_2$ to generate $\mathcal{P} = \{p_1, p_2, \ldots, p_{N_1}, p_{N_1+1}, p_{N_1+2}, \ldots, p_{N_1+N_2}\}$. We use spherical linear interpolation and linear interpolation to interpolate n rotation quaternions and translation matrixes between two camera poses p_i and p_{i+1} by

$$Q_{ij} = \frac{\sin((1-\Delta_j)\theta)}{\sin(\theta)} Q_i + \frac{\sin(\Delta_j \theta)}{\sin(\theta)} Q_{i+1}, \tag{1}$$

$$T_{ij} = (1-\Delta_j) T_i + \Delta_j T_{i+1}, \tag{2}$$

where $\Delta_j = \frac{j}{n+1}, j = 1, \ldots, n$, and θ is the angle between Q_i and Q_{i+1}. Thus, we can get two novel camera poses $\tilde{\mathcal{P}}_1 = \{p_1^1, p_{11}^1, \ldots, p_{1n}^1, p_2^1, \ldots, p_{N_1}^1\}$ and $\tilde{\mathcal{P}}_2 = \{p_1^2, p_{11}^2, \ldots, p_{1n}^2, p_2^2, \ldots, p_{N_2}^2\}$. Note that we discard the interpolation poses between p_{N_1} and p_{N_1+1} in $\mathcal{P}$ for later steps.

B. Change detection with inconsistent IDs

Given $\tilde{\mathcal{P}}_1$, we use the pre-trained 4D GS representation Ψ to generate two image sets $\tilde{\mathcal{S}}_1 = \{\mathbf{I}_1^1, \mathbf{I}_{11}^1, \ldots, \mathbf{I}_{1n}^1, \mathbf{I}_2^1, \ldots, \mathbf{I}_{N_1}^1\}$ and its corresponding image set $\tilde{\mathcal{S}}_1' = \{\mathbf{I}_1^2, \mathbf{I}_{11}^2, \ldots, \mathbf{I}_{1n}^2, \mathbf{I}_2^2, \ldots, \mathbf{I}_{N_1}^2\}$ after change. In order to generate a consistent ID for the same instance, we first use SAM [21] to segment each image to obtain the independent instances. Then, we use DEVA [22], a pre-trained zero-shot tracker, to assign IDs to the instances. By comparing the IDs of I_1^1 and I_1^2, we can identify the change instances at the camera pose p_1^1. Similar operations are conducted for the other poses. The masks of the change instances serve as the corresponding change maps, allowing us to obtain the change mask set $\mathcal{M}_1 = \{\mathbf{M}_1^1, \mathbf{M}_{11}^1, \ldots, \mathbf{M}_{1n}^1, \mathbf{M}_2^1, \ldots, \mathbf{M}_{N_1}^1\}$ for $\tilde{\mathcal{P}}_1$. Same operations are conducted on $\tilde{\mathcal{P}}_2$ to generate $\mathcal{M}_2 = \{\mathbf{M}_1^2, \mathbf{M}_{11}^2, \ldots, \mathbf{M}_{1n}^2, \mathbf{M}_2^2, \ldots, \mathbf{M}_{N_2}^2\}$.

C. 4D Gaussian partition

To generate change maps of new viewpoints, it is crucial to identify which Gaussians belong to the change instances. Inspired by Gaussian Grouping [24], we introduce classification encoding, which is a learnable vector with 16 bits.

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The classification of the changed instances and unchanged backgrounds should be consistent across different rendering views. The SH degree of the classification encoding is set to 0 to model its direct-current component. For the classification encoding e, we adopt the same processing method as the pixel color. We calculate the 2D classification feature f_c for each pixel by the weighted sum of the classification encodings that overlap with the pixel in depth order by

$$f_c = \sum_{i \in \mathcal{N}} e_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_i), \tag{3}$$

where $\mathcal{N}$ is the Gaussian distribution set that overlaps with pixels which is ordered according to depth, α_i is the opacity of the i-th Gaussian in the arrangement. Let $\mathbf{F} \in \mathbb{R}^{16 \times H \times W}$ denote the classification feature matrix. Then we can generate the corresponding change map by

$$\hat{\mathbf{M}} = Cov(\mathbf{F}), \tag{4}$$

where $Cov(\cdot)$ is a convolutional layer that transforms $\mathbf{F}$ into a change mask $\hat{\mathbf{M}}$.

D. Loss function

To learn the new 4D Gaussian representation, we use $\mathcal{S}_1 = \{\mathbf{I}_1^1, \mathbf{I}_2^1, \ldots, \mathbf{I}_{N_1}^1\}$ and $\mathcal{S}_2 = \{\mathbf{I}_1^2, \mathbf{I}_2^2, \ldots, \mathbf{I}_{N_2}^2\}$ as inputs and $\hat{\mathcal{M}}_1 = \{\mathbf{M}_1^1, \mathbf{M}_2^1, \ldots, \mathbf{M}_{N_1}^1\}$ and $\hat{\mathcal{M}}_2 = \{\mathbf{M}_1^2, \mathbf{M}_2^2, \ldots, \mathbf{M}_{N_2}^2\}$ as groundtruth change maps. We propose a 2D loss $\mathcal{L}_{2d}$ with binary cross entropy loss to classify each 3D Gaussian by the change mask as

$$\mathcal{L}_{2d} = -\frac{1}{\|\mathbf{M}\|} \sum_{i \in \mathbf{M}} \sum_{c=0}^{1} p(i) log \hat{p}(i), \tag{5}$$

where $\mathbf{M}$ is the change mask from $\mathcal{M}_1$ or $\mathcal{M}_2$. The unsupervised 3D Regularization Loss $\mathcal{L}_{3d}$ is adopted to regularize the learning of classification encoding e_i, which takes advantage of the 3D spatial consistency by enforcing the classification encodings of the top k-nearest 3D Gaussians to be close in their feature distance. $\mathcal{L}_{3d}$ is defined as

$$\begin{aligned} \mathcal{L}_{3d} &= \frac{1}{m} \sum_{j=1}^{m} D_{kl}(\mathcal{G} \| \mathcal{H}) \\ &= \frac{1}{mk} \sum_{j=1}^{m} \sum_{i=1}^{k} \mathcal{F}(e_j) log(\frac{\mathcal{F}(e_i)}{\mathcal{F}(e_j^{'})}), \end{aligned} \tag{6}$$

where $\mathcal{G}$ consists of the sampled classification encoding e of a 3D Gaussian, the set $\mathcal{H} = \{e_1^{'}, e_2^{'}, \cdots, e_k^{'},\}$ represents its k nearest neighbors in 3D space, and $\mathcal{F} = softmax(Cov(e_i))$.

As the general reconstruction problem [25]–[27], we also use L1 loss and total variation loss $\mathcal{L}_{TV}$ to supervise the quality of the rendered image. The total loss is defined as

$$\mathcal{L} = \mathcal{L}_1 + \mathcal{L}_{TV} + \lambda_{2d} \mathcal{L}_{2d} + \lambda_{3d} \mathcal{L}_{3d}, \tag{7}$$

where λ_{2d} and λ_{3d} are two balance coefficients.

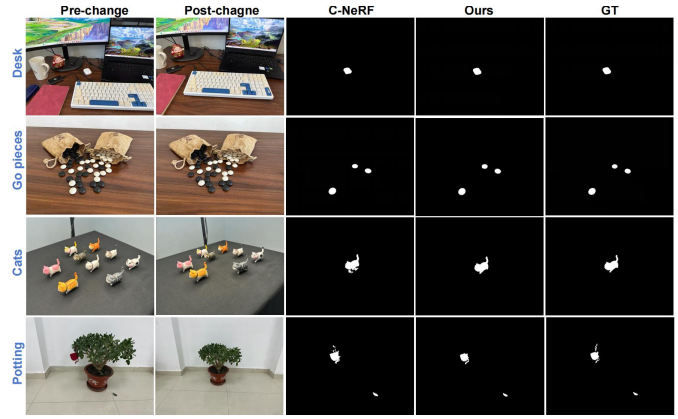

Fig. 2. Examples of CD results of different methods on four scenes. The red arrow indicates the real change.

E. Change detection in a novel view

After the 4D Gaussian trained, we can render an image $\mathbf{I}^t$ as [19] and generate the corresponding change map $\mathbf{M}$ by Eq.4 at the pose p and time t of a novel view.

III. EXPERIMENT

A. Setup

Dataset. We adopt four scenes of C-NeRF [13] to evaluate the performances of different CD methods, including Desk, Chess, Potting, and Cats. Desk consists of 40 training images, Chess has 48 training images, Potting includes 42 training images, Cats is built with 243 training images. Each scene has 10 image pairs for evaluation that are uniformly sampled from different viewpoints.

Baselines. C-NeRF [13] and CYWS-3D [23] are used for comparison. We conduct a quantitative comparison with C-NeRF for it can produce pixel-level prediction. CYWS-3D conducts object-level change detection in two-view and class-agnostic scenarios. The input of CYWS-3D takes two-view images as input and uses depth maps to extract geometric information for change detection, outputting bounding boxes around the changes. To facilitate comparison with CYWS-3D, we use the minimum bounding rectangle of the connected area as the bounding box for our method and C-NeRF.

Criteria. We adopt Precision (P), Recall (R), F1-measure (F_1), and Intersection over Union (IoU) as evaluation metrics.

B. Results analysis

Fig. 2 shows some typical detection results of our method and C-NeRF. Both methods successfully detect the real changes of the four scenes. However, our detected results are more complete than the results of C-NeRF. In the Cat scene, the detected change region of C-NeRF is influenced by the shadows. To demonstrate the robustness of our method, we conduct change detection on two new scenes with wild lighting changes. As shown in Fig. 3, the first scene has non-uniform lighting and the second scene has strong shadows. C-NeRF fails to distinguish the background change from the foreground change whereas our method can detect the changes for both scenes. In Fig. 4, we compare the bounding box results of

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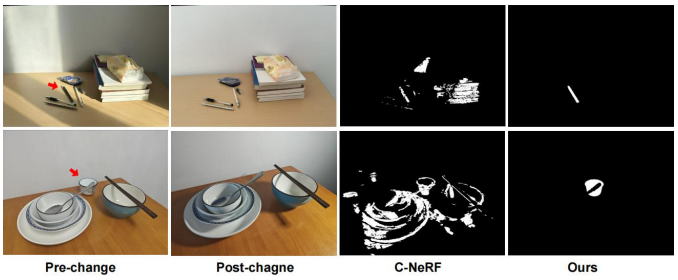

Fig. 3. CD on the scenes with substantial lighting variations. The red arrow indicates the real change.

TABLE I
QUANTITATIVE COMPARISON OF DIFFERENT CD METHODS. THE BEST VALUES ARE HIGHLIGHTED IN **BOLDED**.

Scene	Method	P↑	R↑	F_1 ↑	IoU ↑
Desk	C-NeRF	**96.02**	90.48	**93.17**	**87.24**
	Ours	93.99	**91.88**	92.78	86.75
Chess	C-NeRF	92.66	**99.71**	96.05	92.41
	Ours	**98.64**	93.70	**96.10**	**92.51**
Potting	C-NeRF	71.59	**94.66**	81.75	69.01
	Ours	**93.54**	76.51	**84.31**	**72.67**
Cats	C-NeRF	86.70	90.46	88.54	79.47
	Ours	**97.47**	**93.21**	**92.41**	**91.03**

TABLE II
RUNNING TIME OF MAIN STAGES OF DIFFERENT CD METHODS.

Stage	NeRF/4DGS	CD	Rendering/Gaussian partition	Total
C-NeRF	16.07h	0.75h	0.78h	17.6h
Ours	1.17h	0.1h	1.2h	2.47h

three CD methods. CYWS-3D can detect changes of larger objects, such as the cat (scene of Cats) and bags (scene of Potting). However, it performs poorly when detecting small objects or objects with minimal geometric changes.

We report the quantitative results of our method and C-NeRF in Table. I. The F_1 and IoU values of our method are better than C-NeRF in three scenes. In Table. II, we compare the running time of the different stages on Potting. Note that our method is more efficient than C-NeRF.

C. Discussion

3D Gaussian contains five attributes, including position $x \in \mathbb{R}^3$, scale factor $s \in \mathbb{R}^3$, rotation quaternion $r \in \mathbb{R}^3$, opacity $\alpha \in \mathbb{R}$, and color defined by spherical harmonic(SH) $c \in \mathbb{R}^k$ (where k represents nums of SH functions). 4D Gaussian Splatting uses separate MLPs to compute the deformations of the position Δx, rotation Δr, and scale Δs, to represent both the Gaussian motion and deformation. A direct solution for CD is using a threshold on $(\Delta x, \Delta r, \Delta s)$ to find the changed Gaussians, and then render the change maps. However, as shown in Fig. 5, we can observe that this simple solution cannot generate the desired results. There are two main reasons: (1) Since $(\Delta x, \Delta r, \Delta s)$ change collaboratively, we cannot accurately distinguish a Gaussian into changed or unchanged; (2) A Gaussian might be used to represent multiple objects or backgrounds. The changed Gaussian not only renders the changed objects, but also renders unchanged objects or backgrounds. In contrast, our method does not need to manually set any threshold and can directly generate the change map according to the classification feature matrix.

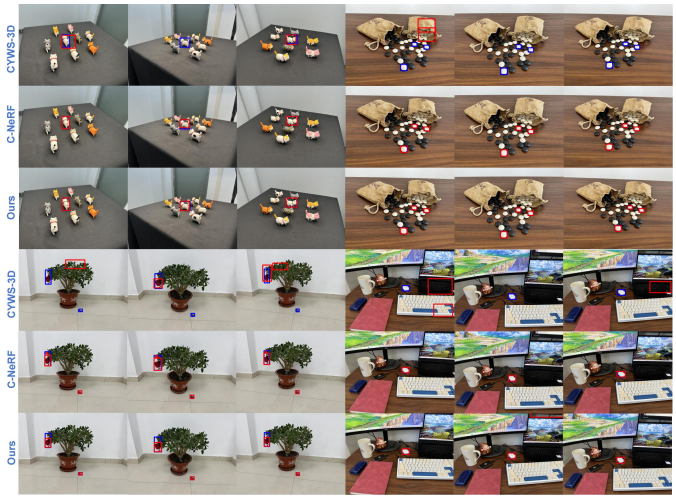

Fig. 4. Comparison of CYWS-3D, C-NeRF, and our method on different scenes. Red boxes are the predictions and blue boxes are the groundtruths.

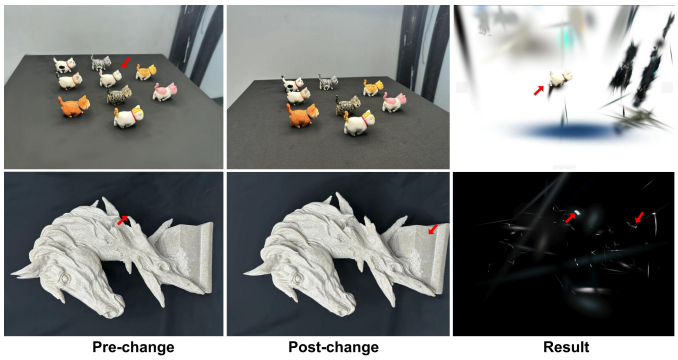

Fig. 5. The CD results obtained by filtering Gaussian distributions with $(\Delta x, \Delta r, \Delta s)$. The red arrow indicates the real change.

IV. CONCLUSION

In this paper, we have introduced a novel approach to 3D instance-level change detection, tackling the challenge of identifying changes in real-world scenarios without relying on controlled conditions or labeled image pairs. Our method utilizes 4D Gaussian Splatting to embed two sets of images into Gaussian representations, enabling the rendering of two coherent image sequences. By segmenting each image and assigning a unique identifier to each instance, we can efficiently detect changed instances through ID comparison. We use the change maps to partition the Gaussians into changed and unchanged by adding a classification encoding to each Gaussian. The change map of any given view direction can be obtained by rendering. Extensive experiments on various instance-level change detection datasets demonstrate that our approach significantly outperforms existing methods like C-NeRF and CYWS-3D, particularly in scenarios with large lighting variations.

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